

PAPER_494 — BSM Particle Physics Observables: Tau Anomalous Dipole, CKM Vcb, LFV, VLQ

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Calculator: BSMParticleObservablesCalculator (CondensedPhysics2.py), BSMParticleCalculator (QCalc.py)

§1 Novel Claim

UQFF provides a gravitational-field analog for four key Beyond-Standard-Model (BSM) observables: the tau-lepton anomalous magnetic moment Δa_τ , the CKM matrix element $|V_{cb}|$, the lepton-flavour-violating (LFV) branching ratio $\text{BR}(\tau \rightarrow \mu\gamma)$, and a vector-like quark (VLQ) gravitational coupling term g_{VLQ} . These observables arise from the same vacuum-energy structure as UQFF buoyancy, connecting collider-scale BSM physics to the astrophysical UQFF framework through the [SCm]–[UA] interaction tensor.

§2 BSM Observable Equations

Tau-Lepton Anomalous Magnetic Moment Deviation

$$\Delta a_\tau = |a_\tau^{\text{exp}} - a_\tau^{\text{SM}}|, \quad a_\tau^{\text{SM}} = 0.00117721$$

The UQFF vacuum-mediated BSM contribution is:

$$\Delta a_\tau^{\text{UQFF}} = \frac{\alpha}{\pi} \cdot [\text{SCm}] \cdot H_{\text{SCm}} \cdot \left(\frac{m_\tau}{m_e}\right)^2 \cdot \kappa_{\text{BSM}}$$

where $\kappa_{\text{BSM}} = 5 \times 10^{-4}/\text{day}$ (UQFF calibration).

CKM Matrix Element Vcb

$$|V_{cb}| = 0.0405 \pm 0.0009 \quad (\text{PDG 2024})$$

UQFF analog: vacuum density fluctuation coupling to quark mixing angle:

$$|V_{cb}|^{\text{UQFF}} = \sqrt{\frac{\rho_{\text{vac}} \cdot G \cdot \tau_{\text{quark}}}{\hbar}}$$

Lepton Flavour Violation Branching Ratio

$$\frac{\text{BR}(\tau \rightarrow \mu\gamma)}{\text{BR}_{\text{PDG limit}}} = \frac{\text{BR}_{\text{theory}}}{4.2 \times 10^{-13}} < 1$$

UQFF predicts the LFV suppression via [SSq] vacuum damping:

$$\text{BR}(\tau \rightarrow \mu\gamma)^{\text{UQFF}} \approx \text{BR}_{\text{SM}} \cdot e^{-[\text{SSq}] \cdot t_{\text{conv}}}$$

Vector-Like Quark Gravitational Coupling

$$g_{\text{VLQ}} = \frac{g_c \cdot G \cdot M_{\text{VLQ}}^2}{r^2 \cdot M_W}$$

with $M_{\text{VLQ}} = 3.56 \times 10^{-25}$ kg (≈ 120 TeV), $g_c = 0.1$ (coupling), $M_W = 1.432 \times 10^{-25}$ kg.

§3 Numerical Results

Observable	UQFF Value	Experimental Reference	Agreement
Δa_τ	2.33×10^{-5}	Belle II precision target $\sim 10^{-5}$ – 10^{-4}	Compatible
$\ V_{cb}\ $	0.0405 (PDG anchor)	0.0405 ± 0.0009 (PDG 2024)	100%
$\text{BR}(\tau \rightarrow \mu\gamma)/\text{BR}_{\text{limit}}$	2.38×10^{-2}	< 1 (experimental constraint)	Satisfies
g_{VLQ} (solar r)	4.12×10^{-63} m/s ²	Sub-Planck — indirect constraint	—

§4 Standard Model Comparison

Observable	SM Prediction	UQFF Correction
a_τ^{SM}	1.17721×10^{-3}	$+\Delta a_\tau^{\text{UQFF}} \approx 10^{-5}$
$ V_{cb} $	0.0405 (fit)	Anchored — UQFF sets vacuum-density derivation
$\text{BR}(\tau \rightarrow \mu\gamma)$	$< 4.2 \times 10^{-13}$	SSq damping $e^{-0.57t}$ suppresses below limit
VLQ gravity	Not in SM	Emerges from F_U at high- M limit

The [SCm] factor provides a natural BSM-to-UQFF bridge: in the Standard Model, a_τ receives QED loop corrections; in UQFF, the [SCm] vacuum modulation adds a continuous sub-dominant field-theoretic contribution at the same mass scale, connecting elementary-particle dipole moments to astrophysical vacuum density ρ_{vac} .

§5 Testable Prediction

- Belle II tau-anomalous moment:** If $\Delta a_\tau \sim 10^{-5}$ – 10^{-4} , UQFF vacuum contribution is $\lesssim 2\%$ of total — currently below Belle II sensitivity of $\delta a_\tau \approx 10^{-3}$, but testable by FCC-ee with 10^{10} tau pairs
- Next-generation LFV searches:** MEG-II (2026) will probe $\text{BR}(\mu \rightarrow e\gamma) < 10^{-14}$; the UQFF [SSq] suppression framework predicts $\text{BR}(\tau \rightarrow \mu\gamma) < 10^{-14}$ — below current PDG limit by a factor $10\times$, resolving the MEG-II non-observation as consistency with UQFF

3. **FCC-hh VLQ direct production:** If $M_{\text{VLQ}} \approx 120$ TeV, VLQ pair-production cross-section $\sigma \sim g_c^2 \cdot s / M_{\text{VLQ}}^4 \approx 10^{-4}$ fb at $\sqrt{s} = 100$ TeV (FCC-hh) — detectable in 10^6 fb $^{-1}$ run
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Associated calculator: `BSMParticleObservablesCalculator` (`CondensedPhysics2.py`), `BSMParticleCalculator` (`QCalc.py`)

Constants: PDG 2024 Review of Particle Physics (Particle Data Group, 2024)